SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Purpose and Scope

This Software Requirements Specification (SRS) defines the complete requirements for TrackWIL version 1.0, the system built and tested in this project. Chapter 3 summarises these requirements, Chapter 4 shows how they were designed and implemented, and Chapter 5 traces every requirement to its test cases in the Requirements Traceability Matrix.

TrackWIL is a web-based system that replaces the paper, PDF and email process for Work-Integrated Learning (WIL) logbooks at CTU Training Solutions. Students record daily work and evidence, workplace mentors verify each entry through a one-time link without needing an account, and coordinators and lecturers monitor every student's progress on a shared dashboard.

In scope (version 1.0): student sign-up and sign-in; daily logbook entries with optional evidence; submission for verification through a one-time link; mentor approval or rejection; entry statuses and verified-hour totals; the coordinator and lecturer dashboard; an SQLite database.

Out of scope (version 1.0): mentor accounts, written mentor feedback, notifications, editing or deleting entries, reports and exports, integration with CTU systems, a mobile application and the use of real personal data.

Table 1: Definitions

Term	Meaning
WIL	Work-Integrated Learning: the workplace component of a qualification
Entry	One day of WIL work recorded by a student (date, description, hours)
Verification link / token	A one-time URL containing a 48-character random token that lets a mentor verify one entry without signing in
Verified hours	The total hours of a student's entries with the status Verified
Overdue	A student with no verified entry in the last 14 days and no entry awaiting verification

2. Users and Stakeholders

Table 2: User classes

User class	Description	Access	Main tasks
Student	Enrolled CTU student completing WIL hours	Account (student)	Sign up, record entries, upload evidence, request verification, track hours
Workplace mentor	Staff member at the host organisation who supervises the student	One-time link only	Review and approve or reject an entry
Coordinator	CTU staff member who manages WIL compliance	Account (coordinator)	Monitor all students' status and hours
Lecturer	CTU lecturer who supports students academically	Account (lecturer)	Monitor all students' status and hours

Secondary stakeholders are the training institution (needs reliable records and POPIA compliance), the development team (needs clear, testable requirements) and the project supervisor (checks alignment with the project objectives and rubric).

3. Functional Requirements

Table 3: Functional requirements with acceptance criteria

ID	Requirement	Priority	Acceptance criterion
FR-01	Sign in with email and password and go to the dashboard for the user's role.	High	Correct details open /student for students and /coordinator for coordinators and lecturers; wrong details show "Invalid email or password." and no session is created. Email matching ignores letter case.
FR-02	Student sign-up with a CTU student email (eight digits @ctucareer.co.za) and a strong password.	Medium	Password needs 8+ characters with upper case, lower case, a number and a special character; a non-CTU or duplicate email is rejected with a message; a valid sign-up signs the student in.
FR-03	Role-based access to pages and actions.	High	Signed-out visitors are sent to the sign-in page; coordinators and lecturers cannot open the student area; students cannot open the dashboard; a student cannot view another student's entry.
FR-04	Create a daily logbook entry.	High	Work date, description and hours are required; hours must be more than 0 and at most 24; the entry is saved as Draft and opened.
FR-05	Attach one optional evidence file to an entry.	Medium	PDF, PNG, JPG, JPEG and WEBP files are accepted and stored under a random name; any other type is rejected with "Evidence must be a PDF or image file."
FR-06	Submit an entry for verification.	High	Only Draft or Rejected entries with a mentor email can be submitted; a 48-character token valid for 7 days is created; the entry becomes Pending verification and the link is shown with a Copy button.
FR-07	Mentor verification through the link without an account.	High	The link shows the student's name, work date, hours, description and status, with Approve and Reject buttons; Approve sets Verified and records the time; Reject sets Rejected.
FR-08	One-time, expiring links.	High	A used link shows "This link was already used"; an expired or unknown link shows an error; neither changes the entry; two clicks on Approve record only one decision.

ID	Requirement	Priority	Acceptance criterion
FR-09	Show entry status.	High	Each entry shows Draft, Pending verification, Verified or Rejected on the logbook and entry pages.
FR-10	Logbook history and hour progress.	High	The logbook lists the student's entries newest first and shows "X of Y verified hours completed".
FR-11	Coordinator and lecturer monitoring dashboard.	High	Every student is listed with On track, Awaiting verification or Overdue, verified hours against required hours, number of entries and number pending.
FR-12	Automatic timestamps.	Medium	created_at, updated_at and verified_at are stored on entries; used_at and decision are stored on links.
FR-13	Protected evidence download.	High	The student who uploaded a file, coordinators and lecturers can open it; another student receives 403 Forbidden and a signed-out visitor receives 401 Unauthorized.
FR-14	Sign out.	High	Sign out removes the session cookie and returns to the sign-in page; protected pages then redirect to sign-in.

4. Non-Functional Requirements

Table 4: Non-functional requirements

ID	Category	Requirement	Measure
NFR-01	Usability	A first-time student can create and submit an entry for verification in under 3 minutes without help.	Timed task (Chapter 5)
NFR-02	Performance	Pages load and actions complete within 2 seconds on the test computer with the demonstration data.	Browser timing
NFR-03	Security	Salted scrypt password hashes; HttpOnly session cookie; tokens with at least 128 bits of randomness; server-side role checks.	Code review and security tests
NFR-04	Reliability and integrity	Database constraints reject invalid data; multi-step updates are atomic transactions.	Constraint tests
NFR-05	Maintainability	TypeScript, layered modules, lint passes, README setup in five commands or fewer.	Lint and README walkthrough
NFR-06	Compatibility	Works in current Chrome and Edge; usable at 375 px and 1280 px wide.	Browser and responsive tests
NFR-07	Privacy	Only data needed for WIL tracking is stored; synthetic data only until written consent.	Data review

5. Use Cases / User Stories

Table 5: Use cases

ID	Use case	Actor(s)	Requirements
UC-01	Sign in / sign up	Student, coordinator, lecturer	FR-01, FR-02, FR-03
UC-02	Create logbook entry	Student	FR-04, FR-09, FR-12
UC-03	Upload evidence (extends UC-02)	Student	FR-05, FR-13
UC-04	Submit for verification	Student	FR-06
UC-05	Approve or reject via link	Workplace mentor	FR-07, FR-08
UC-06	View monitoring dashboard	Coordinator, lecturer	FR-10, FR-11
UC-07	Sign out	Student, coordinator, lecturer	FR-14

The user stories US-01 to US-08 and their acceptance criteria are listed in Chapter 3, Section 3.7.

6. Data Requirements

TrackWIL stores its data in an SQLite database with three tables. The full data dictionary is in the Database Design document.

- **users:** id, full name, email (unique, case-insensitive), role (student, coordinator or lecturer), password hash and required hours.
- **logbook_entries:** id, student id (foreign key to users), work date, description, hours (more than 0 and at most 24), evidence file name, status, mentor email and the created, updated and verified timestamps.
- **verification_tokens:** token (primary key), entry id (foreign key to logbook_entries), expiry time, time used and decision (approved or rejected).

Validation rules: every rule is checked in the server action and, where possible, again by a database constraint (foreign key, UNIQUE or CHECK), so that invalid data cannot be stored even if a check in the code is missed.

Evidence files are stored in the data/uploads folder under random names; only the file name is stored in the database.

Retention: demonstration data is synthetic and can be reset at any time with npm run db:reset. If real data is used after permission is granted, personal information will be deleted when the project result is finalised, unless the campus sets another period.

7. Interfaces

Table 6: User interface screens

Screen	Route	Users
Sign in	/	All account holders
Sign up	/signup	New students
My logbook	/student	Students

Screen	Route	Users
New entry	/student/entries/new	Students
Entry details and verification link	/student/entries/[id]	Students
Mentor verification	/verify/[token]	Workplace mentors (no sign-in)
Coordinator dashboard	/coordinator	Coordinators and lecturers

Software interfaces: the application uses Node.js's built-in node:sqlite module to access the SQLite database file data/trackwil.db, and the server file system for evidence files. No external services are called.

Communication interfaces: browsers communicate with the server over HTTP (HTTPS in a hosted deployment). Forms call Next.js Server Actions; evidence files are served by the route /api/evidence/[fileName].

8. Constraints and Assumptions

- **Time:** the system had to be built, tested and demonstrated by 29 October 2026, so the scope was limited to the core workflow.
- **Technology:** Next.js 16 with TypeScript and Tailwind CSS; SQLite through Node.js 22.13 or newer; Git and GitHub for version control.
- **Legal and privacy:** POPIA applies; no real personal information is collected before written campus permission, so all testing uses synthetic data.
- **Deployment:** the prototype runs locally; cloud hosting is out of scope.
- **Assumption:** users have a modern web browser and an internet or network connection to the server.
- **Assumption:** students send the verification link to the correct mentor, and mentors check the work before approving it.
- **Assumption:** each student needs 200 verified hours (the default required hours for new accounts).

9. Acceptance Criteria

The solution is accepted when:

1. Every High-priority functional requirement (FR-01, FR-03, FR-04, FR-06 to FR-11, FR-13 and FR-14) passes its test cases in the Test Cases and Results workbook.
2. A student can complete the full journey (sign in, create an entry with evidence, submit it, mentor approves through the link, hours increase) during the live demonstration using synthetic data.
3. Coordinators and lecturers can identify students who are on track, awaiting verification or overdue from the dashboard.
4. Security tests confirm that role restrictions, one-time links and protected evidence downloads work, and any open security defects are documented with a planned fix.
5. All defects found during testing are logged in the Defect Log and either fixed and retested or documented as known limitations.